



T(6th Sm.)-Computer Sc.-H/(DSE-B-3)/CBCS

2021

COMPUTER SCIENCE — HONOURS

Paper : DSE-B-3

(Introduction to Computational Intelligence)

Full Marks : 50

The figures in the margin indicate full marks.

*Candidates are required to give their answers in their own words
as far as practicable.*

Answer **question no. 1** and **any four** from the rest.

1. Answer **any five** questions : 2×5
 - (a) What do you understand by state space search?
 - (b) Differentiate data driven search and goal driven search.
 - (c) Define fuzzy set.
 - (d) Discuss any two characteristics of Prolog.
 - (e) What is Heuristic function?
 - (f) What do you mean by Semantic Net?
 - (g) Distinguish between Machine learning and Deep learning.
 - (h) Discuss in brief about fuzzy relationship.
2.
 - (a) Discuss with an example about the Depth First Search algorithm.
 - (b) Discuss about the time complexity of Depth First Search. (4+4)+2
3. Write short notes on (**any two**) : 5×2
 - (a) Gradient Descent Method
 - (b) Feed forward neural network
 - (c) De-fuzzification
 - (d) Training dataset.
4.
 - (a) Differentiate between probabilistic approach and fuzzy logic based approach.
 - (b) What is the sequence of steps taken in designing a fuzzy logic machine? 5+5
5.
 - (a) What do you understand by Game playing in Artificial Intelligence?
 - (b) Explain Minimax search algorithm for game playing. 3+7

Please Turn Over

6. (a) Differentiate fuzzy set and crisp set.

(b) Prove that $(\tilde{A} \cap \tilde{B})^c = \tilde{A}^c \cup \tilde{B}^c$

where, $\tilde{A} = \{(x_1, 0.4), (x_2, 0.3)\}$

and $\tilde{B} = \{(x_1, 0.2), (x_2, 0.6)\}$

and $\tilde{A}$ and $\tilde{B}$ are fuzzy set.

5+5

7. $\tilde{A}$ and $\tilde{B}$ are two fuzzy sets.

$$\mu_{\tilde{A}}(x) = \frac{x}{x+1} \text{ and } \mu_{\tilde{B}}(x) = \frac{1}{x}$$

Find membership function of each of the following :

(a) $\tilde{A}^c$ and $\tilde{B}^c$

(b) $\tilde{A} \cup \tilde{B}$

(c) $\tilde{A} \cap \tilde{B}$

(d) $(\tilde{A} \cup \tilde{B})^c$

(e) $(\tilde{A} \cap \tilde{B})^c$

2×5

8. (a) Discuss with an example about A^* algorithm.

(b) Is iterative deeping search complete?— Justify your answer.

7+3
